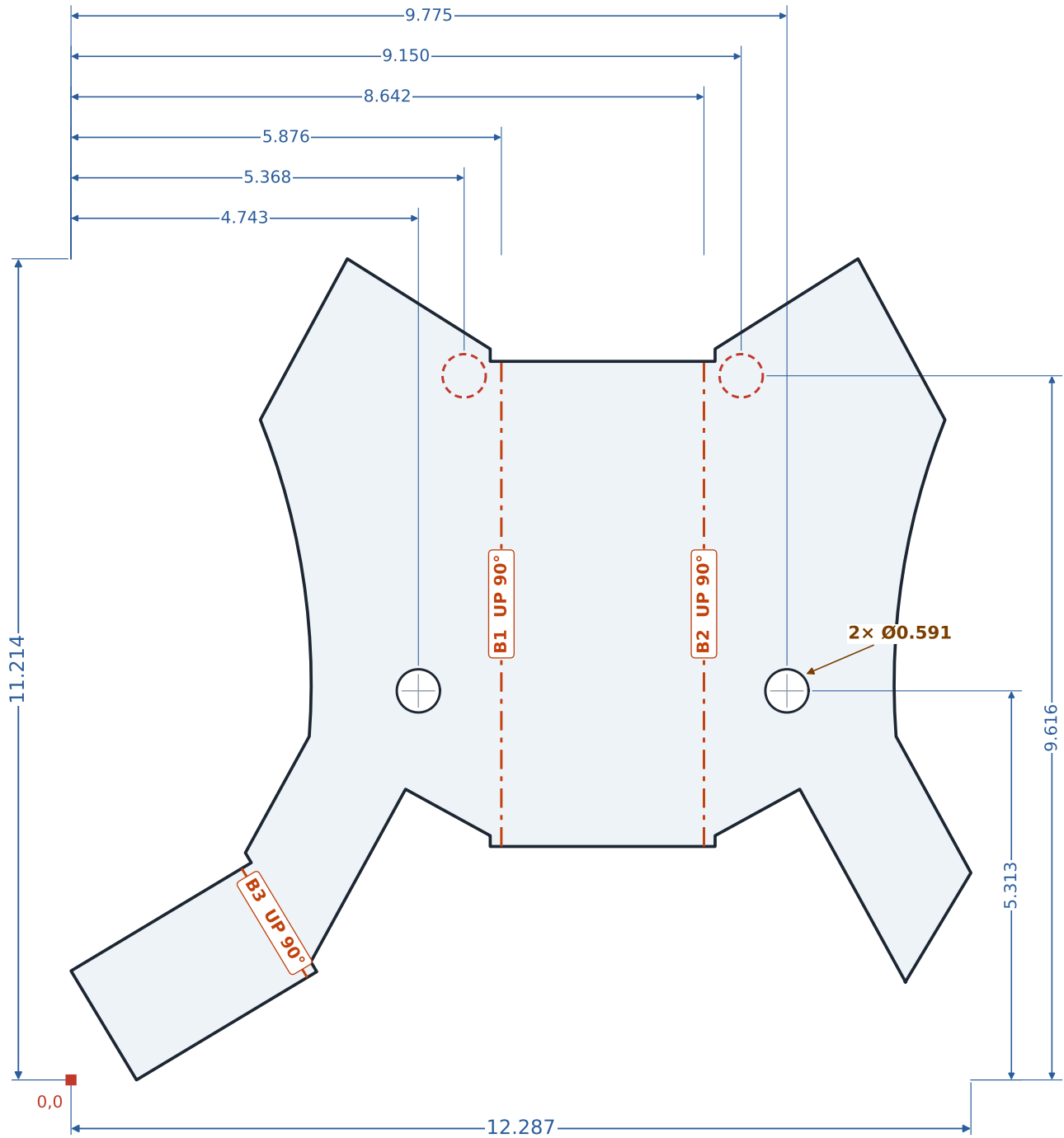


FR-10-1 · Part 1 "U-body" — 2 sides + wall A-H + flange E-F (3 bends)

10 ga 0.135" (3.42 mm) · HR steel sheet A1011 CS · flat blank 12.287" × 11.214" · ≈ 2.55 lb each · QTY 1 FR + 1 FL = 2 per vehicle

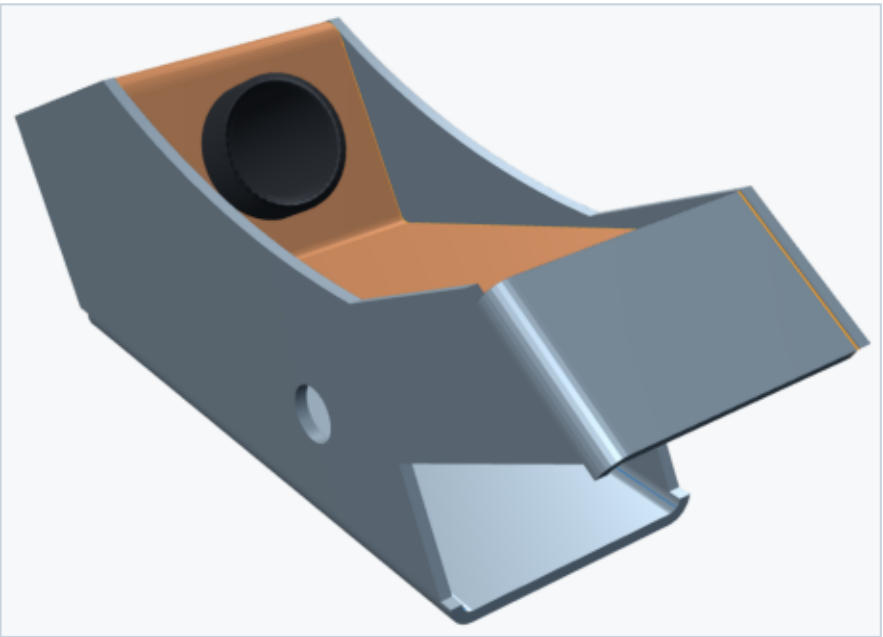
FLAT PATTERN (as laser-cut, marked side up)



NOTES

- 1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - 2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - 3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - 4. Deburr all edges, remove dross.
 - 5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.
- * bends 90° up, R = t; the 2 Ø0.591 holes near A (½" from A-B, ⅝" from A-H) are not in the flat — drill after bending

FORMED PART (after bending)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	UP (toward)	R=t 0.135	2	X = 5.876
B2	90.0°	UP (toward)	R=t 0.135	3	X = 8.642
B3	90.0°	UP (toward)	R=t 0.135	1	(2.331, 2.888)→(3.225, 1.399)

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø0.591	4.743	5.313	laser
H2	Ø0.591	9.775	5.313	laser
D1	Ø0.591	5.368	9.616	DRILL LATER
D2	Ø0.591	9.150	9.616	DRILL LATER

FILES

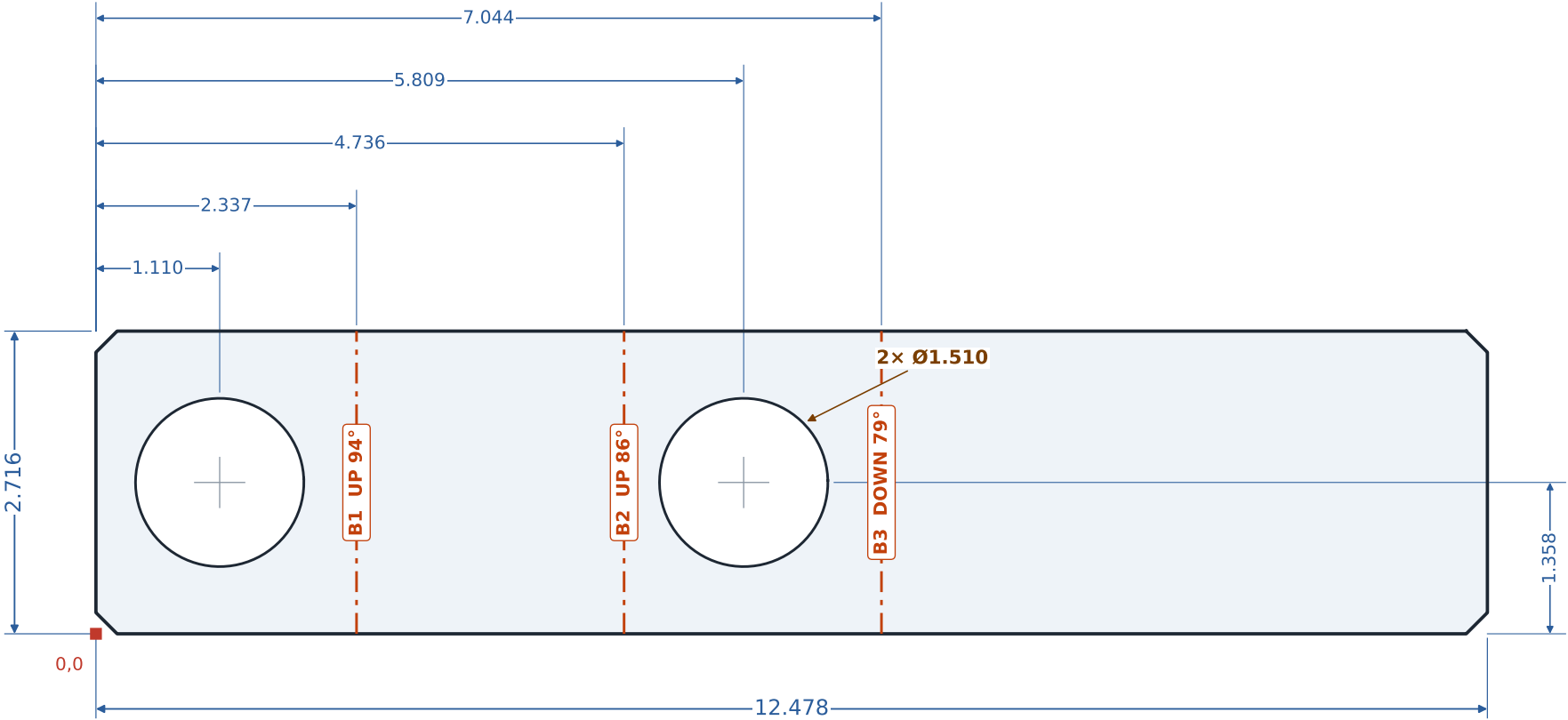
- FR-10-1_FR_cut.dxf (right, qty 1 / vehicle)
- FR-10-1_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-10-1_ref.dxf (reference: + bend lines, drill-later)

FR-10	Bracket: U-body + strip + tube	HR steel sheet A1011 CS 10 ga 0.135" (3.42 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FR-10-1 · Part 1 "U-body" — 2 sides + wall A-H + flange E-F (3 bends)		QTY 1 FR + 1 FL = 2 per vehicle	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 1/3

FR-10-2 · Part 2 "strip" — A-B · B-C · shelf · inner plate (3 bends)

10 ga 0.135” (3.42 mm) · HR steel sheet A1011 CS · flat blank 12.478” × 2.716” · ≈ 1.15 lb each · QTY 1 FR + 1 FL = 2 per vehicle

FLAT PATTERN (as laser-cut, marked side up)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	93.9°	UP (toward)	R=t 0.135	5	X = 2.337
B2	86.1°	UP (toward)	R=t 0.135	6	X = 4.736
B3	79.2°	DOWN (away)	R=t 0.135	4	X = 7.044

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø1.510	1.110	1.358	laser
H2	Ø1.510	5.809	1.358	laser

FILES

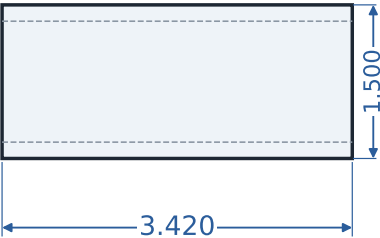
- FR-10-2_FR_cut.dxf (right, qty 1 / vehicle)
- FR-10-2_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-10-2_ref.dxf (reference: + bend lines, drill-later)

NOTES

- Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
 - Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
 - Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
 - Deburr all edges, remove dross.
 - Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.
- * X 79.2° opposite way; B 93.9°, C 86.1° up; corners 3/16"×45°

FR-10	Bracket: U-body + strip + tube	HR steel sheet A1011 CS 10 ga 0.135” (3.42 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
FR-10-2 · Part 2 "strip" — A-B · B-C · shelf · inner plate (3 bends)		QTY 1 FR + 1 FL = 2 per vehicle	Tol.: cut ±0.010” · bend ±1° · IR = t	2026-10-03	SHEET 2/3

FR-10 · tube cut list



Tube OD 1.5 / ID 1.181 — cut 3.42" (through A-B and the shelf)

Stock: DOM round steel tube OD 1.500" × ID 1.181" (wall ≈ 0.160")

Cut length 3.420" ±1/32", square ends, deburr

QTY 1 per side × 2 (FR + FL) = 2 per vehicle

FR-10	Bracket: U-body + strip + tube	see stock above	Units: inch · NOT TO SCALE	Sienna WAV frame	REV A
Tube cut list		see quantities above	Tol.: cut ±0.010" · bend ±1° · IR = t	2026-10-03	SHEET 3/3